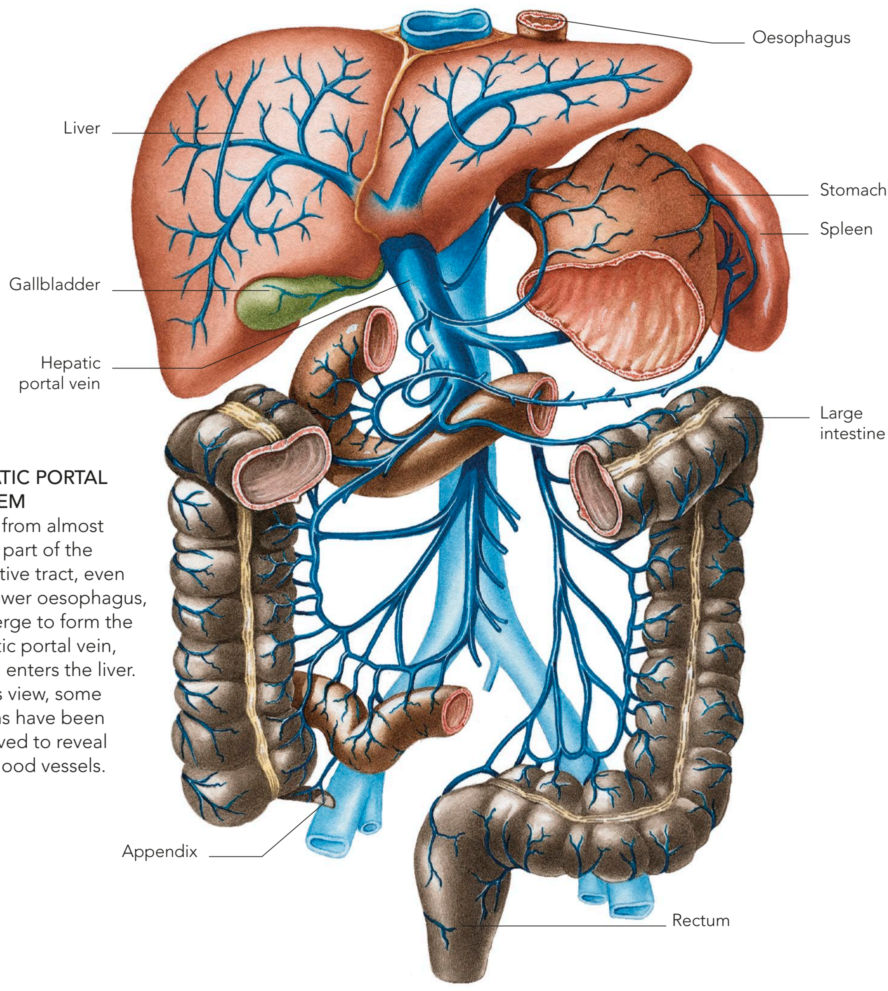
THE HEPATIC PORTAL CIRCULATION

The liver receives two blood supplies. The hepatic artery delivers oxygen-rich blood to the liver. In addition, the hepatic portal vein supplies the liver with the oxygen-poor, nutrient-rich blood that comes from the digestive tract, before this blood returns to the heart and is pumped throughout the body. This hepatic portal circulation enables the liver to stop toxins absorbed in the intestines from reaching the rest of the body. It also helps to regulate the levels of many other substances in the bloodstream. Veins from several organs, including the intestines, pancreas, stomach, and spleen, drain into the hepatic portal vein. It is around 8cm (3in) long and supplies up to four-fifths of the blood into the liver. The flow-rate increases after a meal, but falls during physical activity as blood is diverted from the abdominal organs to skeletal muscles.



HEPATIC PORTAL SYSTEM

Veins from almost every part of the digestive tract, even the lower oesophagus, converge to form the hepatic portal vein, which enters the liver. In this view, some organs have been removed to reveal the blood vessels.